



DALHOUSIE
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Psychology and Neuroscience
CELEBRATING 70 YEARS

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Psychology and Neuroscience 44th Annual Graham Goddard In-House Conference

April 26-27, 2018
Life Sciences Centre
Room 5260



The 2018 Psychology and Neuroscience Graham Goddard In-House Conference Welcomes you!

We would like to thank you for attending the 44th Annual Graham Goddard In-House Conference. This is an excellent opportunity to hear about the ongoing research of students, faculty and colleagues. It is your participation that makes this event a success.

Thanks should also be extended to the In-House Conference Committee. The time that the following individuals put into the planning and organization of the In-House is greatly appreciated: Raymond Klein, Tara Perrot, Yvonne Brandelli, Matthew Orr, Victoria Patterson, Colin Pridy, R.J. Redden, Flint Schwartz and Susan Lowerison.

Dr. Graham V. Goddard (1938-1987) was a leading researcher at Dalhousie University. Goddard invented the kindling model of epilepsy that has, since its introduction in the 1960s, become widely used research tool in studies of long-term potentiation (LTP) and epilepsy. His work was published in Nature, Science, along with other high profile journals. During his time at Dalhousie, Goddard helped establish Dalhousie as a leading research institution. Goddard also organized the annual Psychology Department In-House Conference, to help investigators in the department update others about their work and encourage academic dialogue. It is in his honour that we continue this tradition today.

Friendly Reminders:

Talks will be held in LSC 5260

The new 4th Floor Study Centre (4332) will host refreshments during breaks, lunches on both days and the posters.

All posters will be up during lunch on both days but authors are expected to present their posters on the assigned days as noted in the program.

Supper will be provided in the 2nd floor Atrium Lounge on Thursday and there will be a simple “beer” reception in the Atrium lounge at the end of the conference on Friday.

#	Time	Presenting Author	Title
Thursday AM - 8:30-10:00 AM Chair - Flint Schwartz			
T1	8:30-8:45	Tara Perrot	Psychology & Neuroscience: Where we've been and where we're going
T2	8:45-9:00	Flint Schwartz	Expressions of Distress among Indigenous and Non-Indigenous Adults Living in an Urban Setting: Considering Within-Group Differences according to Acculturation and Enculturation
T3	9:00- 9:15	Toni Spinella	The impact of cannabidiol (CBD) and stimulus expectancy on subjective and physiological responses to stress.
T4	9:15-9:30	Christopher DeWolfe	Gender Differences in Physical Activity are Partially Explained by Anxiety Sensitivity
T5	9:30-9:45	Dylan Miller	Feeding microstructure approaching feeding cessation in an insect: Effects of satiation, immune challenge, and stress.
T6	9:45-10:00	Elizabeth O'Leary	Effects of probiotic treatment on maternal care, with outcomes for adolescent stress resiliency and gut health
10:00 - 10:30 AM Refreshment Break			
Thursday AM: 10:30 AM - 12:00 noon Chair - Colin Pridy			
T7	10:30-10:45	Penny Corkum	Better Nights, Better Days: An accessible and sustainable transdiagnostic intervention?
T8	10:45-11:00	Matt Orr	The influence of role-playing games on perceived social competence.
T9	11:00-11:15	Lydia Musingo	A Meta-analysis of Partner Effects on Alcohol Use
T10	11:15-11:30	Aaron Shephard	Caffeine cue reactivity: The impact of caffeine-related stimuli and expectancies on caffeine craving.
T11	11:30-11:45	Colin Pridy	Music's charms vary by anxiety sensitivity level
T12	11:45-12:00	Richard Drake	Does AFO support for walking reduce dual-task costs in individuals post-stroke?
12:00 - 1:30 PM Lunch/Posters			

#	Time	Presenting Author	Title
P1	Posters and lunch Study Centre LSC4332	Raquel Nogueira-Arjona	Drinking to keep pace: Testing the moderating influences of sensation seeking and extroversion on alcohol consumption in drinking buddy dyads
P2		Ioan Mahu	Specificity of Four Personality Correlates of Concurrent Substance Use Among Opiate Substitution Therapy Clients
P3		Marco Lee	Do I Want Babies Now? Reproductive Investment of Gryllus texensis under the Interaction of Aging and Immune Challenge.
P4		Clara Lownie	An ERP study of reading ability and polyunsaturated fatty acid consumption in childhood
P5		Tristan Park	Investigating the Path from Childhood Maltreatment to Alcohol Problems in a Sample Child Welfare-Involved Youth
P6		Noelle Strickland	Self-compassion as a resilience factor for alcohol-involved sexual assault and negative emotional outcomes.
P7		Emily Puddicombe	What Influences Children’s Sensitivity to Violations of Authority as a Moral Foundation?
P8		Nacera Hanzal	Changes in Anxious Mood, Stress, and Cannabis Use Across the Menstrual Cycle: A Daily Diary Study
P9		Lindsey Puddicombe	Alertness and Human Performance
Thursday PM: 1:30-3:00 PM Chair - Yvonne Brandelli			
T13	1:30-1:45	Meghan Rossi	Partner Perceptions of Women’s Pain Self-Efficacy in Postpartum Dyspareunia: A Longitudinal Dyadic Study
T14	1:45-2:00	Yvonne Brandelli	Parental Perceptions of Pain and Treatment Adherence in Juvenile Idiopathic Arthritis
T15	2:00-2:15	Elizabeth Myles	Patterns and Functions of Non-Suicidal Self-Injury among Adolescents with Concurrent Substance Use
T16	2:15-2:30	Hélène Deacon	Reading and learning in the wild: Young readers learning of the spellings and meanings of words during their independent reading.
T17	2:30-2:45	Alex Ryken	Prosodic Cues in Text: The Missing Link Between Prosodic Sensitivity and Reading Comprehension in Young Readers?

#	Time	Presenting Author	Title
T18	2:45-3:00	Victoria Patterson	A psychometric validation of a developmental psychopathology measure: The Youth Experience Tracker Instrument (YETI)
3:00 - 3:30 PM Refreshment Break			
Thursday PM: 3:30-5:00 PM Chair - Jasmine Aziz			
T19	3:30-3:45	Jasmine Aziz	Role of Working Memory in Visual Search among Older Adults
T20	3:45-4:00	T-Jay Anderson	Effects of Caffeine on Sensory Gating in Schizophrenia
T21	4:00-4:15	Mohammad Habibnezhad	Employing Rescorla’s truly random condition to measure truly exogenous covert attention
T22	5:15-4:30	Jack Solomon	Motor Execution and Imagery are Unique Processes
T23	4:30-4:45	Mohammed Al-Hamdani	The effects of self-affirmation messaging and illness sensitivity on drinking intentions
T24	4:45-5:00	Kayla Joyce	Changes in coping and social motives for drinking and alcohol consumption across the menstrual cycle
5:00 PM Dinner in the Atrium Lounge			
Friday AM: 8:45-10:00 AM - Science Slam Chair - Raymond Klein			
S1	Science Slam 6 min each	Lindsey Power	Brain-Computer Interface Advancements
S2		Abigail Shapiro	Effects of University-Affiliated Alcohol Marketing on Students
S3		Alysha Pelley	The Microbiome and Depression
S4		Nicole Pickett	The Impact of Service Dogs on Quality of Life
S5		Zoë Dworsky-Fried	Returning from a Vegetative State
S6		Joshua Dort	Magic Mushrooms as Medication
S7		Lana Kelly	Physiological and Psychological Effects of Going Offline
S8		Zoë Lazar-Kurz	Morphological Changes Resulting from Pregnancy
S9		Jessie Pappin	Mindfulness and Mental Health
S10		Colin McCormick	The Potential of Probiotics as a Treatment for Anxiety Disorders
S11		Matt Orr	Cyberpsychology and eHealth
10:00 - 10:30 AM Refreshment Break			

#	Time	Presenting Author	Title
Friday AM: 10:30 AM- 12:15 PM Chair - Jennifer Stamp			
T25	10:30-10:45	Jennifer Stamp	Making Science Accessible: A-RAD-ical Approach
T26	10:45-11:00	Dana Lee-Baggley	The Art and Science of Behaviour Change
T27	11:00-11:15	Graham Hood	Pornography Consumption and Attitudes Supportive of Violence against Women in College-aged Males
T28	11:15-11:30	Robin Curtis	Exploring the Effects of Afterlife Beliefs on Generosity and Prudence
T29	11:30-11:45	Emre Fertan	What do we actually measure when “measuring behaviour”?
T30	11:45-12:00	Sean Mackinnon	Everything your undergrad students are (supposed to be) learning about statistics: Part 1
T31	12:00-12:15	John Christie	Everything your undergrad students are (supposed to be) learning about statistics: Part 2
12:15 - 1:30 pm Lunch/Posters			
P10	Posters and lunch Study Centre LSC4332	Jessie Pappin	Investigating the Effect of Visual Noise on Cognitive Load and Memory using Item-Method Directed Forgetting
P11		Zoe Lazar-Kurz	Prism Adaptation in Older Adults: an ERP Analysis
P12		Heather Gerrie	In vivo application of the photoswitch BENAQ in a rat model of acquired retinal degeneration
P13		Hannah MacNeil	Investigating the ability of superoxide to pass dendritic spine plasma membranes in hippocampal neurons via chloride channels using fluorescence imaging
P14		Sydney Harris	The unresolved role of the cell adhesion molecule family, neuroligins, and activity on the stability of synapses between hippocampal neurons
P15		Michaela Purdon	Behavioural phenotypes of the MDGA2 ^{+/-} mouse model of autism spectrum disorder (ASD)
P16		Kristen Lee	Effects of Atrx haploinsufficiency and mother-pup interactions on cognitive, social and emotional behaviours in adulthood
P17		Ray Klein	Using the Hebb Legacy to Integrate Psychological Science

#	Time	Presenting Author	Title
Friday PM: 1:30-3:00 PM Chair - RJ Redden			
T32	1:30-1:45	Tovah Kashetsky	Do songbirds care about human singing?
T33	1:45-2:00	Ralph Redden	Using temporal order judgments to dissociate two inhibitions of return
T34	2:00-2:15	Theresa Gaughan	The Effect of Memory Instruction on Spatial Associations
T35	2:15-2:30	Katie Douglas	Measuring the N400 effect in a naturalistic setting
T36	2:30-2:45	Ray Klein	Testing the networks of attention: From the ANT to the Attention-Trip
T37	2:45-3:00	Alena Galilee	Ortho-semantic learning of novel words in Grade 3 students: An ERP study
3:00 - 3:30 PM Refreshment Break			
Friday PM: 3:30-5:00 pm Chair - Tara Perrot			
T38	3:30-3:45	Steve Shaw	A provisional equivalent circuit for the fly's eye space.
T39	3:45-4:00	Filip Kosel	Characterization of social deficits in the 5xFAD mouse model of Alzheimer's disease
T40	4:00-4:15	Jill King	Modulating GABAergic interneuron activity in mouse primary visual cortex alters response to contrast
T41	5:15-4:30	Richard Brown	Frailty as a measure of healthspan and lifespan in transgenic mice
T42	4:30-4:45	Jessica Pelley	Hubel & Wiesel – Their Contributions to Visual Research
T43	4:45-5:00	John Barresi	Freud's Rat Man And The Meaning Of The Rat Torture
5:00 - ? Reception in the Atrium Lounge			

In-house Abstracts 2018

Talks

T1: Psychology & Neuroscience: Where we've been and where we're going

Author: Tara Perrot

T2: Expressions of Distress among Indigenous and Non-Indigenous Adults Living in an Urban Setting: Considering Within-Group Differences according to Acculturation and Enculturation

Authors: Flint Schwartz and Amy Bombay

Abstract: Despite high levels of reported psychological distress in Indigenous peoples in Canada, research has not explored potential differences in expressions of psychological distress in this diverse population. This study seeks to explore: 1) how Indigenous adults express their distress through somatic versus psychological symptoms relative to non-Indigenous adults; 2) how symptom type relates to single-item mental/physical health self-ratings in both groups; and 3) whether associations vary by cultural orientation (acculturation and/or enculturation). Ongoing participant recruitment involves Indigenous and non-Indigenous adults experiencing symptoms of psychological distress, anxiety, and/or depression within the Halifax Regional Municipality. Measures include the Vancouver Index of Acculturation, the Kessler Psychological Distress Scale, single-item self-rated physical/mental health, and a demographics questionnaire (N=150). A series of regression analyses will assess the research questions and the potential moderation by cultural group, acculturation, and/or enculturation. Implications for more culturally responsive mental health care and health research will be discussed.

T3: The impact of cannabidiol (CBD) and stimulus expectancy on subjective and physiological responses to stress.

Authors: Toni C. Spinella, Sean P. Barrett

Abstract: Despite the increasing availability and upcoming legalization of recreational cannabis use, research findings are mixed on its potential risks and benefits. Additionally, the mechanisms underlying the potential therapeutic effects of cannabidiol (CBD) remain unclear. This study seeks to examine the independent and combined influences of CBD-related expectation and pharmacology on stress response. Adult participants will be recruited from the Halifax community. Prior to stress induction participants will be randomly assigned to one of four conditions in a balanced placebo design, where the content and instructions regarding the content of the oral pill will vary between participants. Subjective ratings of stress will be collected, and heart rate and cortisol levels will be assessed before and after the stress induction paradigm. Findings from this study will provide insight into how CBD impacts subjective and physiological responses to stress, a critical step towards informing future research, clinical practice, and public health policy.

T4: Gender Differences in Physical Activity are Partially Explained by Anxiety Sensitivity

Authors: Christopher EJ DeWolfe, Margo C Watt, Pablo Romero-Sanchiz, Sherry H Stewart

Abstract: Objectives - Women engage in less physical activity than men. Given that women have greater anxiety sensitivity (i.e., fear of arousal-based body sensations) and anxiety sensitivity is inversely related to physical activity participation, this study sought to determine if anxiety sensitivity mediates gender differences in physical activity. Method - A sample of 802 undergraduates completed the Anxiety Sensitivity Index-3 (Taylor et al., 2007) and a Lifestyles Questionnaire assessing physical activity levels. Results - A significant indirect effect of gender on physical activity via anxiety sensitivity was shown, 5.56 , $SE = 2.81$, $p < .05$, $95\% CI [1.31, 12.78]$, $P_M = .0843$. Women engaged in significantly less physical activity and had significantly greater anxiety sensitivity.

Anxiety sensitivity was significantly and inversely related to physical activity. Conclusions - Results suggest that anxiety sensitivity partially explains gender differences in physical activity. Anxiety sensitivity reduction interventions might increase physical activity participation and reduce the existing gender gap.

T5: Feeding microstructure approaching feeding cessation in an insect: Effects of satiation, immune challenge, and stress.

Authors: DW Miller, SA Adamo

Abstract: As animals approach satiation, their motivation to feed decreases. However, there are a variety of factors other than satiation that can affect feeding motivation. The tobacco hornworm caterpillar, *Manduca sexta*, is an ideal system for studying feeding motivational physiology. It shows no circadian rhythm in its feeding and has a relatively simple nervous system. It reduces feeding when ill, exposed to model predators, and during certain developmental periods. The force with which an animal bites during feeding has been linked to feeding motivation in many other species. By noninvasively measuring the force applied to food during feeding, changes in *M. sexta* motivation can be examined across a feeding period and between different conditions. Understanding how this caterpillar terminates a self-generated behaviour such as feeding, provides insight into the regulation of motivation.

T6: Effects of probiotic treatment on maternal care, with outcomes for adolescent stress resiliency and gut health

Authors Elizabeth O'Leary

Abstract: Prenatal and early postnatal events are important for shaping our development, including the stress response. The mechanisms by which early life events shape both the brain and stress response involve maternal care behaviour provided to offspring. The connection between the brain and intestinal tract, the "gut-brain axis", has also demonstrated modulatory effects on stress responding. The administration of probiotics in pregnancy, or early life has demonstrated to provide protective effects for early life stress, resulting in positive outcomes for offspring. To date, there has been no investigation into the effects of probiotics on maternal care behaviour quality, and subsequent offspring outcomes, such as anxiety-like behaviours, corticosterone levels, and gut health. To investigate these effects, we propose the following research aims: a) investigate differences in maternal care behaviours due to probiotic administration, and gut health outcomes in dams/offspring and b) investigate offspring anxiety-like behaviours/stress responding, and biomarkers of stress and inflammation.

T7: Better Nights, Better Days: An accessible and sustainable transdiagnostic intervention?

Authors: Penny Corkum & the BNBD Team

Abstract: Introduction: Behavioural insomnia in children is a common and impairing problem. Providing intervention in an accessible manner is critical. As such, we have been building and evaluating a suite of eHealth Interventions, called Better Nights, Better Days (BNBD). **Methods:** We are currently conducting a Canada-wide randomized control trial (RCT) for parents of typically developing children (BNBD-TD), have conducted four foundational studies to inform the development of BNBD for children with neurodevelopmental disorders (BNBD-NDD) and conducted focus groups to inform the development of a version for adolescents and young adults (BNBD-Youth). **Results:** While the RCT is still in progress, the results from the foundational studies have provided strong support for BNBD. **Conclusions:** BNBD shows promise as an accessible and effective eHealth intervention across ages and diagnoses. Sustainability of this program will be discussed.

T8: The influence of role-playing games on perceived social competence.

Author: Matthew Orr

Abstract: This study examined the relation between tabletop role-playing games and social competence in young adults. A mixed methods approach was used to extend previous research in this area. The Strengths and Difficulties Questionnaire was used to assess perceived social competence from participants and semi-structured interviews were conducted to collect detailed qualitative data about experiences with role-playing games. Six young adults (3 men, 3 women) between the ages of 19 and 24 ($M = 21.2$, $SD = 2.6$) participated. Responses on the SDQ indicated that people who play role-playing games have typical strengths and difficulties with respect to social competence. The qualitative description indicated the emergence of four themes: Content Focus, Social Focus, Creativity Focus, and Identity Focus. These themes are consistent with findings from previous case study research, in that role-playing games may have the potential to improve social competence.

T9: A Meta-analysis of Partner Effects on Alcohol Use

Authors: Lydia Muyingo, Sherry Stewart, Simon Sherry

Abstract: The Partner Influence Hypothesis postulates that one partner's alcohol use (AU) influences the other's AU over time. While studies have documented romantic partner influences on drinking, results about the magnitude and sex-specific nature of such effects remains unclear. A meta-analysis (MA) is proposed to clarify the existence of partner influences on drinking in romantic couples, provide a more definite answer about the magnitude of such effects and to examine sex differences in the extent of partner influence. A comprehensive search will be conducted across six databases. Studies that have measured both members of a couple's AU across at least two time-points will be included. Data will be analyzed using Actor-Partner Interdependence Modelling and pooled effect sizes, and effect moderators will be examined. The results of this study will clarify the existence and strength of partner influences on AU. Moreover, the proposed MA could demonstrate how interpersonal relationships affect risky behavior.

T10: Caffeine cue reactivity: The impact of caffeine-related stimuli and expectancies on caffeine craving.

Authors: Aaron Shephard, Carol Morrissey, Sean Barrett

Abstract: Caffeine is consumed by approximately 90% of adult, yet its potentially addictive properties have been understudied. Cue reactivity is an observable physiological and subjective response that can be used to measure a drug's addictive potential. Although cue reactivity is documented with many different substances, including cocaine and tobacco, no previous studies have examined this effect with caffeine. This study aims to address gap. Participants will be adult daily coffee drinkers. Following 12-hour caffeine abstinence, participants' caffeine craving will be assessed. They will then receive either caffeine-containing or placebo gum and be given either accurate or inaccurate information regarding the gum's caffeine content. Following an absorption period, they will be exposed to neutral and coffee related stimuli, before having their level of craving measured again. It is hypothesized that coffee cues will induce craving in participants and that this will occur regardless of actual or perceived caffeine consumption.

T11: Music's charms vary by anxiety sensitivity level

Authors: Colin B. Pridy, Margo C. Watt, Pablo Romero-Sanchiz, Christopher J. Lively, and Sherry H. Stewart

Abstract: Music's many uses include mood- and arousal-regulation. Anxiety sensitivity (AS; fear of arousal-related somatic sensations) is a risk factor for anxiety-related disorders. This study examined relations between AS dimensions (physical, cognitive, social concerns) and music-listening motivations. Undergraduates ($N = 817$; 77.8% female; $M_{age} = 19.06$, $SD_{age} = 2.71$) completed the Anxiety Sensitivity Index-3 and three validated music-listening motive measures. An exploratory

factor analysis (EFA) was conducted to determine factors underlying these various music-listening motive measures. Six underlying music-listening motives: coping, revitalization, connection, social enhancement, conformity, and sensory-motor. Regressions tested the relation of each AS dimension to these music-listening motive dimensions. AS cognitive concerns uniquely predicted coping, connection, and conformity music-listening motives while AS social concerns uniquely predicted coping and revitalization music-listening motives. Results add music-listening to the growing list of behaviours (e.g., alcohol use, smoking) linked to higher AS that are aimed at arousal- and negative mood-regulation.

T12: Does AFO support for walking reduce dual-task costs in individuals post-stroke?

Authors: Richard S. Drake, Kim Parker, Kerry Clifton, James Adderson, Anita Mountain, Gail A. Eskes

Abstract: Cognitive-motor interference, as measured by dual-task walking (performing a mental task while walking), is common with hemiplegia due to stroke. Ankle-foot orthoses (AFOs) are custom-made splint devices that provide stability to the foot and ankle and help prevent foot drop, a gait deficit common after stroke. AFO use has been shown to improve gait parameters such as speed and step time, which are often negatively impacted by dual-task walking. We investigated the protective effect of AFOs (i.e., whether AFOs could reduce cognitive-motor interference) in dual-task walking in 21 individuals with post-stroke hemiplegia using a 2 (with/without AFO) by 2 (with/without dual task) repeated-measures design. The results suggest that the use of an AFO improves gait overall (e.g., improved step length), particularly while dual-tasking, although its effect on gait variability (e.g., stride regularity) is less straightforward. Theoretical and clinical implications of dual-task effects during walking will be discussed.

T13: Partner Perceptions of Women's Pain Self-Efficacy in Postpartum Dyspareunia: A Longitudinal Dyadic Study

Authors: Meghan Rossi, Kathleen Merwin, & Natalie Rosen

Abstract: Childbirth is a significant risk factor for the development of dyspareunia (pain during intercourse). Women with postpartum dyspareunia that are higher in pain self-efficacy (PSE) may be more motivated to seek out methods of reducing pain and exhibit more adaptive cognitive-affective reactions to pain. Partner perceptions of women's greater PSE have been linked to women's lower pain during intercourse. Prior research suggests that partners can be biased in their perceptions of the patient's pain experience, which may have consequences for the couples' well-being. This study aims to examine whether partner perceptions of women's PSE are biased at 3 months postpartum and whether this bias predicts women's pain and functional impairments in both members of the couple at 6 months postpartum. Data will be analyzed according to the Truth and Bias Model of Judgment via Response Surface Analysis. Findings may improve our understanding of how couples manage pain after childbirth.

T14: Parental Perceptions of Pain and Treatment Adherence in Juvenile Idiopathic Arthritis

Authors: Yvonne Brandelli, Christine Chambers, Adam Huber, Jennifer Stinson, Jennifer Parker, Perri Tutelman, Jennifer Wilson

Abstract: Juvenile Idiopathic Arthritis (JIA), a disease characterized by pain and swollen joints, affects over 24,000 children across Canada. While pharmacological and non-pharmacological treatments aim to manage the pain and prevent irreversible joint damage, adherence is often a challenge. This study will examine the parents' role in treatment adherence. It is hypothesized that there will be less treatment adherence in families where parents demonstrate pain catastrophizing and fears of pain, and that medication, side effects, and levels of pain will moderate these relationships, and barriers to treatment will mediate these relationships. One-hundred-and-fifty caregivers of children under the age of 18 with JIA will be recruited via social media to complete a

T5: Patterns and Functions of Non-Suicidal Self-Injury among Adolescents with Concurrent Substance Use

Abstract: In adolescents who engage in non-suicidal self-injury (NSSI), the prevalence of comorbid substance use can be as high as 60%. The co-occurrence of both behaviours in the same individual increases the risk of harmful consequences. The purpose of the current study was to examine adolescents exhibiting non-suicidal self-injury (with or without concurrent substance use) and compare the principal motives underlying its use. A total of 53 participants ($M_{age} = 15.23$, $SD = 1.59$), who underwent psychiatric assessment at the Child and Youth Addiction and Mental Health Services, between 2012 and 2015, completed a series of relevant questionnaires. Significant differences in intrapersonal reinforcement functions between the unitary NSSI and the comorbid groups, specifically with respect to positive reinforcement motives, were found. Based on these results, the nature of self-injurious behaviour may be experienced differently in individuals who also engage in substance use.

Abstract: We explore whether younger readers can learn the spellings and meanings of words through their independent reading. A total of 66 Grade 1 and 2 English-speaking children read stories containing novel words referring to new inventions (e.g., a veap, used to clean fish tanks). Following on this independent reading, children in both grades were above chance in choosing correct spellings and meanings for the new words. We also found that individual differences in skill in learning the spellings of novel words were related to word reading. In contrast, individual differences in skill in learning the meanings of the novel words were related to specifically to reading comprehension. These findings suggest that independent reading is a means through which children can acquire rich information about the meanings and spellings of new words and that children's skill in this learning might support reading skills in general.

Abstract: Reading comprehension, or the ability to understand the meaning of text, is one of the most important skills children learn in school. Understanding how various metalinguistic skills support reading comprehension is essential to improving literacy curriculum. One skill which is related to reading comprehension in young readers is prosodic sensitivity, which is an awareness of the rhythmic cues in speech. However, the reason for this relationship is unclear. One possibility is that young readers are aware of how prosodic cues are represented in text, e.g. through punctuation. The proposed research will test the possibility that awareness of how prosodic cues are represented in text mediates the relationship between prosodic sensitivity and reading comprehension in young readers. This will be done by testing 150 grade 3-5 students individually in schools and examining their reading comprehension, prosodic sensitivity, and awareness of how prosodic cues are represented in text.

T18: A psychometric validation of a developmental psychopathology measure: The Youth Experience Tracker Instrument (YETI)

Authors: VC Patterson, A Pencer, B Pavlova, A Awadia, A Zwicker, E Howes Vallis, N Campbell, V Drobinin, LE MacKenzie, R Uher

Abstract: Severe mental illness (SMI) is defined as any major psychotic or affective disorder. Antecedents, early manifestations of risk for SMI, may allow for targeted intervention to prevent the onset of SMI. However, there are no brief self-report measures that screen for antecedents in youth. The Youth Experience Tracker Instrument (YETI) measures six antecedents: affective lability, anxiety, basic symptoms, psychotic-like experiences, insomnia, and depressive symptoms. This study aims to establish the psychometric properties, factor structure, and optimal cut-off points of the YETI. The sample was 208 offspring of parents with and without SMI aged 8 to 17-years participating in the FORBOW study. A confirmatory factor analysis procedure was used to establish the questionnaire's structure. Receiver operating curves were used to select a cut-off point for the scale. This study's findings will validate a new instrument that may facilitate detection of youth at high risk for the development of SMI.

T19: Role of Working Memory in Visual Search among Older Adults

Authors: Jasmine R. Aziz, Raymond M. Klein, Gail A. Eskes

Abstract: Visual search is a goal-directed process whereby we scan for and locate important objects (targets) in our environment, from choosing products on shelves at the grocery store to scanning for pedestrians while driving. Working memory involves temporarily holding and/or manipulating information in mind. During visual search, working memory helps us to remember target features and previously searched locations. Visual search and working memory abilities both decline with age, which can interfere with daily activities (e.g., shopping, driving). This study will examine the role of working memory in visual search in older adults. We will manipulate working memory demands by varying whether a picture of the target (target template) is displayed during the search. We will relate this template effect to search performance, working memory abilities, and how frequently participants look at the template when displayed. Findings may assist in developing strategies promoting working memory and visual search abilities in aging.

T20: Effects of Caffeine on Sensory Gating in Schizophrenia

Authors: T Anderson, K McKearney, P Tibbo, D Fisher

Abstract: A predominant symptom of schizophrenia (SZ) is a deficit in sensory gating, or an inability to selectively filter out extraneous stimuli. Sensory gating also has deep connections to neurological functioning and is a large component of alerting effects of caffeine consumption. On average, individuals with SZ consume nearly double the norm of caffeine compared to typical populations (predicted self-medication). The EEG-derived event-related potential P50 is a robust index of sensory gating and provides the high temporal resolution needed to investigate this early automatic process. 15 SZ patients and 15 healthy controls were given caffeine and placebo treatments in two EEG recording sessions. Participants were recorded passively listening auditory click stimuli (typical populations suppress attentional response). P50 amelioration from caffeine was only seen in patients and not healthy controls. Indicating that a dose of 250 mg of caffeine was sufficient to ameliorate P50 indexed sensory gating in patients but not controls.

T21: Employing Rescorla's truly random condition to measure truly exogenous covert attention

Authors: Mohammad Habibnezhad, Michael A Lawrence and Raymond M Klein

Abstract: Attention prioritizes processing of some information over the other. Attention can be engaged either exogenously (by an external event e.g. a loud noise) or endogenously (intention to focus on a task e.g. reading). Because endogenous control is thought to be engaged when stimulus

properties (such as spatial location) can be predicted, measuring purely exogenous orienting requires removing such predictability. Following this guide, previous studies of exogenous spatial orienting have used peripheral cues that were **spatially uninformative** about the upcoming target's location. However, these cues were almost always **temporally informative** about the upcoming target's time of presentation and this may have elicited some degree of endogenous control. We employed Rescorla's truly random procedure to eliminate the temporal information communicated by cue and target contingency. Forty participants were recruited to complete a variation of Posner's cueing paradigm, in which cues and targets were presented temporally randomly in one of two equiprobable peripheral locations. The results demonstrated a biphasic pattern of early facilitation and late inhibition (IOR). To evaluate the effect of removing temporal contingencies, we intend to implement the next experiment using fixed SOAs, mixed SOAs and non-aging paradigm.

T22: Motor Execution and Imagery are Unique Processes

Authors: Jack P. Solomon, Sarah N. Kraeutner, Timothy Bardouille, Shaun G. Boe

Abstract: Motor Imagery (MI), the mental rehearsal of movement, is a method of motor learning that has been thought to parallel motor execution (ME). The neural mechanism(s) underlying the primary difference between MI and ME, the lack of movement in MI, is not well understood. Here we generated evidence from neuroimaging data to test two theories that exist to explain inhibition of movement in MI. We hypothesized MI and ME would diverge at the early stages of performance. Eighteen participants performed a grasping task using both MI and ME while we recorded their brain activity to ascertain common measures of brain activity that relate to movement planning and execution. Our results show that in the frequency domain, MI and ME differ early in the preparation phase for movement but the field strength only differed post-movement. This refutes our hypothesis but, still challenges the assertion that MI and ME are parallel processes.

T23: The effects of self-affirmation messaging and illness sensitivity on drinking intentions

Authors: M Al-Hamdani, S Smith, SH Stewart

Abstract: Little is known about the direct/indirect effects of self-affirmation alcoholic beverage labels, or illness sensitivity, on intentions to reduce drinking. Two studies addressed these gaps. Experiment 1 examined the effect of self-affirmation labels on alcohol appeal and intentions to reduce drinking. Labels were designed to increase attention to harms followed by an attempt to maintain self-integrity. Experiment 2 compared two models of the effects of self-affirmation labels and illness sensitivity on drinking-reduction intentions: direct vs. indirect via negative affect. In Experiment 1, self-affirmation reduced alcohol appeal and increased intentions to reduce drinking. In Experiment 2, negative affect mediated effects of illness sensitivity/self-affirmation on drinking-reduction intentions and the direct and indirect models were equally plausible. Results support the use of self-affirmation messaging in alcohol labelling practice to deter problematic drinking. Results also emphasize the importance of negative affect as a mechanism through which self-affirmation/injury sensitivity increase intentions to reduce drinking.

T24: Changes in coping and social motives for drinking and alcohol consumption across the menstrual cycle

Authors: Kayla Joyce, Amanda Hudson, Roisin O'Connor, Kara Thompson, Megan Hodgins, Tara Perrot, & Sherry Stewart

Abstract: Alcohol use increases during the (pre-)menstrual (characterized by negative affect) and ovulatory (characterized by positive affect) phases of the menstrual cycle (MC), suggesting women may drink for particular emotion-focused reasons during specific phases. Ninety-four normally-cycling women completed daily diary surveys, examining state drinking motives and alcohol consumed across an entire MC. Women reported increases in drinking to cope premenstrually, peaking menstrually, and declining mid-cycle. Women reported a slight increasing trend across the

cycle in social motives, while enhancement motives remained stable across the cycle. Drinking to cope predicted greater alcohol quantities menstrually. While social motives predicted greater alcohol quantities during the follicular and ovulatory phases, enhancement motives were unrelated to drinking quantity. Results indicate that clinicians should be attentive to MC phase when treating reproductive-aged women with alcohol use disorders (AUD). Findings may help develop new, and improve existing, interventions for women with/at risk of, comorbid mood and AUD.

T25: Making Science Accessible: A-RAD-ical Approach

Author: Jennifer Stamp

Abstract: Dissemination of scientific research findings from the community of experts to the broader public is challenging, especially since the source literature makes for intimidating reading. Therefore people turn to headlines or websites for information, which often provide oversimplified or erroneous explanations. The **A**ccessible **R**esearch **A**rticle **D**atabase (**ARAD**) is a collaboration between faculty from Dalhousie University and Science Atlantic, with the aim of building a collection of easy-to-understand, primary research articles suitable for undergraduate teaching. Contributions come from course instructors based upon two criteria: 1) article is relatively short and 2) contributor deems it to be accessible to an undergraduate science student. Then novice science students evaluate the articles for difficulty, identify jargon, and record reading time. To date, there are over 90 evaluated articles for Psychology and Neuroscience, organized by topic based on introductory textbook chapters, and housed in a shareable Google spreadsheet. The project is expanding into other scientific disciplines (Biology and Physics) and the creation of a searchable database.

T26: The Art and Science of Behaviour Change

Authors: Dayna Lee-Baggley

Abstract: Dr. Dayna Lee-Baggley, Assistant Professor in Faculty of Medicine, is a new cross-appointee in the Department of Psychology and Neuroscience. She will provide an overview of her clinical work as a health psychologist and her research on behaviour change. She has an active research program on chronic disease, behavior change, obesity, professional resiliency and Acceptance and Commitment Therapy. Clinical and research opportunities for students will be reviewed.

T27: Pornography Consumption and Attitudes Supportive of Violence against Women in College-aged Males

Authors: Graham Hood and Tim Juckes

Abstract: College-aged males consume pornography often (>4-5 a week) and for a considerable duration (>30 minutes at a time). We correlated consumption and type of pornography with attitudes towards women. We tested 28 male undergraduates (18-26 years old, mean age 22) using the Dalhousie Pornographic Typology and Consumption Scale (DPTCS). We collected consumption habits (i.e., frequency, duration, frequency of masturbation, internet search preferences, and consumption preferences) and viewing preferences based on 33 search terms (e.g., Softcore, Submission, Amateur Lesbians) commonly found on pornography streaming sites (e.g., Pornhub.com). Some search terms target DREVA pornography (Degrading, Rough, Extreme, Violent, and Abusive). Participants used the 33 search terms in an extremeness card sort and ratings weighted consumption scores. Weighted consumption was correlated with attitude scores on the Illinois Rape Myth Acceptance Scale and the Attitudes Supportive of Violence Scale. Results showed a positive correlation between the card sort and attitudes ($r = 0.35$). However, consumption habits negatively correlated with attitudes ($r = -0.42$), and DREVA-only consumption correlated negatively with attitudes ($r = -0.33$). Results suggest college-aged males recognize extremeness but consumption

is not detrimental to their attitudes toward women. This is a pilot study; results should be interpreted with caution.

T28: Exploring the Effects of Afterlife Beliefs on Generosity and Prudence

Author: Robin Curtis

Abstract: Priming thoughts of supernatural monitoring and punishment has been shown to increase prosocial behaviour among certain religious adherents. This study investigated whether priming thoughts of the afterlife might have similar effect. 80 Christians and 80 atheists were recruited from the undergraduate population. Participants completed a survey, including a writing prime, in which half described their expected situation after graduation, and half described their expected situation after death. Preliminary results in pre- and post-prime generosity and prudence measures suggest that contemplating the afterlife may facilitate generosity toward strangers among Christians, while inhibiting money-saving behaviour among atheists. Implications and follow-up directions are discussed.

T29: What do we actually measure when “measuring behaviour”?

Authors: Emre Fertan, Aimee A. Wong, Richard E. Brown

Abstract: Behavioural tests are commonly used in neuroscience to study the function of brain regions, validate animal models of neurological diseases, and test therapeutics. One common problem with these behavioural tests is that researchers often choose the easiest test to measure a specific behaviour and fail to take any external factors into account when interpreting the results. In this talk, I will provide three examples from our laboratory which demonstrates that test results may not always show what we think they do.

1. In the triple test of anxiety, 5xFAD mice are not less anxious, as found in other studies, but they are avoiding the closed arms of the EPM.
2. In the Hebb-Williams maze female 3xTg-AD mice have low food motivation, which results in poor cognitive performance.
3. By treating glaucoma in DBA mice, we showed that spatial learning in the MWM is confounded by vision deficits.

T30: Everything your undergrad students are (supposed to be) learning about statistics: Part 1

Author: Sean Mackinnon

Summary: Statistical Methods I (PSYO 2501) is a mandatory course for our psychology and neuroscience majors, and is intended to be students' first statistics course with an emphasis on techniques used in our discipline. Since 2016, the curriculum has been revised and refocused. The present talk will focus on these changes to facilitate communication about what are students are (and are not) taught in this important pre-requisite with an eye to aligning material with topics covered in upper-level courses. In general, the revisions to the class involve three key changes: (a) covering fewer statistical techniques, but spending more time on each topic; (b) deemphasizing arithmetic hand-calculations in favor of conceptual, interpretation, and applied problems; and (c) introducing active learning workshops with real-time support from TAs and instructors. Notable omissions from the curriculum include factorial ANOVA and repeated-measures ANOVA. A few descriptive statistics on student performance will also be presented.

T31: Everything your undergrad students are (supposed to be) learning about statistics: Part 2

Authors: John Christie

Abstract: Statistical Methods II (PSYO 3502) is a mandatory course for our psychology and neuroscience honours students. In 2012 the course was heavily revised with incremental improvements ongoing. It focuses on understanding the edge cases ($p = .051$), thinking meta-analytically, figuring out how to analyze data on your own, and reading results sections critically,

including checking to see if authors both back up what they say, and making logically coherent arguments. It builds on the simulation techniques that were introduced in 2501 but has the students generate the simulation themselves. It does not build a large library of statistical techniques. It expands from 2501 with repeated measures designs, factorial designs, and logistic regression. A small amount of the time is spent at the end of the course demonstrating multi-level designs and model comparison.

T32: Do songbirds care about human singing?

Authors: Tovah Kashetsky, Nafisah Ali, Will Brien, and Leslie Phillmore

Abstract: In songbirds, the immediate-early gene ZENK is expressed in neural regions critical for auditory processing after playback of acoustic stimuli. Expression levels change with breeding condition or sex of the bird and complexity of stimuli, but most studies find that own-species, or conspecific, vocalizations produce the most ZENK immunoreactivity (Zenk-ir) compared to vocalizations of other species (heterospecific), white noise, pure tones, or silence. In this study we asked whether stimuli considered heterospecific to a songbird, the zebra finch, could induce higher-than-expected Zenk-ir if the stimuli were made meaningful to the bird. To do this, we quantified Zenk-ir (and another immediate-early gene c-fos) in response to playback of human vocalizations the birds had learned to associate with reward or punishment in an operant conditioning paradigm. Compared to birds that had never heard these stimuli before, operant consequences appear to increase Zenk, but not c-fos response to heterospecific stimuli, meaning these regions are important for processing meaningful, not necessarily conspecific, stimuli.

T33: Using temporal order judgments to dissociate two inhibitions of return

Authors: Ralph S Redden, Austin J Hurst & Raymond M Klein

Abstract: Early work on inhibition of return (IOR) using a temporal order judgment (TOJ) task showed no effect on arrival time judgments, suggesting this inhibitory effect is acting at a post-perceptual information processing stage. Recent work, however, suggests that there are two forms of IOR; one affecting the quality of inputs and the other affecting responding, whereby the type of effect that is manifest is contingent upon the activation state of the reflexive oculomotor system. We tested this theory in a TOJ task, where subjects were required to either make a prosaccade (output form) or antisaccade (input form) after the onset a spatially uninformative peripheral cue, and subsequently execute a TOJ response. We found inhibited perceptual processing when the input form was generated, but no effect on TOJs in the group that elicited the output form. These findings provide converging evidence that there are two forms of IOR: an input effect operating on a saliency map, and an output effect operating on a priority map.

T34: The Effect of Memory Instruction on Spatial Associations

Authors: T Gaughan, & TL Taylor

Abstract: In a directed forgetting paradigm, participants are presented with a series of words and instructed to remember or forget each word. When instructed to forget, attention is withdrawn from the word as well as the location the word was presented in. Certain words have associations to specific locations in space, such as “sky” having an “up” association. The present study examined the effect memory instruction has on spatial associations. Participants were presented with an item-method directed forgetting paradigm. Words had an association to upper or lower space, or no spatial association. Following an instruction to remember or forget each word, participants localized a target that appeared above, below, or at center and response times were collected. A subsequent memory test revealed a directed forgetting effect: participants recognized more words that were instructed to-be-remembered than words instructed to-be-forgotten. Contrary to the literature, response time analyses revealed no indication that memory instruction has an effect on congruency.

T35: Measuring the N400 effect in a naturalistic setting**Authors:** Caitriona (Katie) Douglas, Antoine Tremblay, Aaron Newman

Abstract: Recently, neuroscientists researching social cognition have begun using a two-way EEG technique called hyperscanning to record the brain activity from two participants simultaneously to better understand naturalistic social interactions. Researchers have demonstrated that it is possible to detect ERPs using hyperscanning in a social joint-task scenario (Loehr et al., 2015). Given that much of psycholinguistic research involves studying ERPs, we developed a hyperscanning method to detect ERPs in a naturalistic conversational setting. We recruited pairs of participants to have a scripted conversation using a script that contained specific amounts of low and high frequency nouns as stimuli because a frequency-modulated N400 effect had been previously reported in the literature (Rugg 1990; Hendrix et al. 2017; Van Petten et al. 2014). We successfully detected the N400 ERP in a conversational setting using hyperscanning, but did not find a significant difference in amplitude between the low and high frequency conditions.

T36: Testing the networks of attention: From the ANT to the Attention-Trip**Authors:** Raymond M Klein, Colin McCormick, Swasti Arora & Sana Rehan

Abstract: Even though W. James asserted that “everyone knows what attention is” the term, attention, is not a single, simple concept. Rather, attention refers to several basic neuro-cognitive processes. Perhaps the most influential taxonomy of attention, according to which there are three isolable networks of attention (alerting, orienting and executive control), has been advanced by Posner and colleagues. The Attention Network Test (ANT) was developed to serve as a quick and easy to administer assay of these networks. The ANT has been widely used to measure the efficacy of these three attention networks. World-wide and local interest in this test will be described and some limitations will be discussed. Our development of a more engaging version, that we call the Attention-Trip, will be presented along with some recent data we have collected using this game-like task.

T37: Ortho-semantic learning of novel words in Grade 3 students: An ERP study**Authors:** Alena Galilee, Lisa Beck, Catherine Mimeau, S. Hélène Deacon, Aaron J Newman

Abstract: It has been proposed that reading development field should move from examining *established* knowledge of orthography and semantics to assessing children’s *capacity to learn* the meaning of both spelling and meaning of new words, referred as ortho-semantic learning (Deacon, 2012). Our main objective was to examine how event-related potentials (ERP) change with ortho-semantic learning. Thirteen native English speaking students in Grade 3 learned novel word spellings and meanings by reading short stories. Immediately after learning, participants completed a lexical decision task in which they were presented with real words, novel words from the stories, non-words, consonant strings, and false fonts. Two different ERP components were investigated: N170 (word recognition) and N400 (lexical semantics). Our preliminary results suggest that after brief exposure, the ortho-semantically novel words were processed similarly to real words. Additional analyses will examine relationships between individual differences in word reading, ortho-semantic learning, and the ERPs.

T38: A provisional equivalent circuit for the fly’s eye space.**Author:** Steve Shaw

Abstract: Why elements of the fly’s visual electroretinogram (ERG) occur with the sizes and polarities that they do remains unexplained, because the equivalent circuit of the eye’s extracellular space is not understood. Exploring this involves locating where extracellular barrier(s) are formed, and demonstrating how both photocurrents and microelectrode currents escape from the eye. Measured resistance depth profiles and dye diffusion point to an extracellular resistance barrier (R_{barrier}) in and around the lamina synaptic zone. The conventional suggested current path runs from

the retina through the optic lobe, but results here point to a new route, currents escaping via the cornea. This pathway has a 3–5 fold lower resistance (R_{cornea}) than the deep route through the optic lobe ($R_{\text{barrier}} + R_{\text{deep}}$). Responses are measured relative to a reference electrode in the haemolymph, but this ground reference is necessarily interposed between R_{deep} and R_{cornea} . The resulting equivalent circuit dictates that circulating photocurrents produce positive voltages around the lamina synapse, which then invert negative out into the retina. This correctly predicts the observed polarity of ERG responses from the photoreceptors and some other sources.

T39: Characterization of social deficits in the 5xFAD mouse model of Alzheimer's disease

Authors: K Kosel, P Torres Munoz, JR Yang & TB Franklin

Abstract: Alzheimer's disease is a progressive neurodegenerative disorder that is characterized by a number of cognitive and behavioural effects; while memory loss is the most well known, patients also present with a number of neuropsychiatric symptoms, including depressive symptoms, anxiety, irritability, agitation, and aggression. This study examines whether a transgenic mouse model of Alzheimer's disease (5xFAD) that exhibits learning and memory deficits can also be used to model social changes in humans with the disease. We tested 5xFAD mice in behavioural tests for olfactory perception, responses to odour cues, sociability, social novelty preference, and free social interaction. Results indicate that transgenic female mice may exhibit altered processing and/or responses to social cues compared to wild-type controls, and that transgenic males may exhibit increased aggression compared to wild-type controls. This suggests that the 5xFAD strain may provide a useful model of social deficits present in Alzheimer's disease patients.

T40: Modulating GABAergic interneuron activity in mouse primary visual cortex alters response to contrast

Authors: Jillian L King, Cheryl Gill, Jared T Shapiro, Nathan A Crowder

Abstract: The visual system has long been used to study cortical information processing. The genetic tools available in mice have allowed researchers to examine the roles of specific types of neurons in the primary visual cortex (V1) with a level of control that was previously inconceivable. Studies that have optogenetically modulated the activity of parvalbumin (Pvb+) and somatostatin (Sst+) expressing GABAergic interneurons indicate that these two populations may counteract the activity of pyramidal neurons (Pyr) differently, although there have been contradictory findings. Vasoactive intestinal peptide (VIP+) expressing GABAergic interneurons have been under-examined because it is believed that they have a negligible role. In the current study, we combined single-unit electrophysiology with optogenetic modulation of either Pvb+, Sst+ or VIP+ interneurons to provide additional data to help resolve the contradictory findings regarding Pvb+ and Sst+, and to determine whether VIP+ interneurons have a role in Pyr responsivity.

T41: Frailty as a measure of healthspan and lifespan in transgenic mice

Authors: Richard E Brown, Alice E Kane, Aimee A. Wong, Susan E Howlett,

Abstract: Mouse models of AD have differences in life expectancy. Frailty is a measure of aging which is independent of chronological age and can be used to compare age-related changes in mice with different lifespans. This study used the mouse frailty index (FI) which is based on deficit accumulation to explore genotype and sex differences in lifespan and healthspan of 3xTg-AD mice. Male 3xTg-AD mice aged 300-600 days had a higher FI score than either male wild-type or female 3xTg-AD mice, and the elevated frailty scores were accompanied by parallel increases in mortality. Frailty increased exponentially with age and higher rates of deficit accumulation elevated mortality risk in all groups. When mice were stratified by FI score, frailty predicted mortality, at least in females. The mouse clinical FI provides a valuable tool that can be used to evaluate healthspan in mice of different strains and in transgenic mice.

T42: Hubel & Wiesel – Their Contributions to Visual Research**Author:** Jessica Pelley

Abstract: David H. Hubel and Torsten Wiesel were pioneers in studying the function and plasticity of the visual system. In 1958, at Johns Hopkins University, they began a collaboration that would continue for over 20 years. Their seminal work, including the 1959 paper “Receptive fields of single neurones in the cat’s striate cortex” was some of the first research demonstrating that simple stimuli could have profound effects on the brain and behaviour. These findings ultimately led to the 1981 Nobel Prize in Physiology or Medicine “*for their discoveries concerning information processing in the visual system*”. Their influences have extended to the Dalhousie community, influencing the research conducted by Donald Mitchell, Kevin Duffy and Nathan Crowder concerning the visual system. In 1998, David H. Hubel was awarded an honorary degree from Dalhousie University, and visited the medical school in 2004 as a Picchione Scholar to speak about his research on the influences of the environment on early-life brain development. Overall, the research of Hubel and Wiesel was crucial in developing the field of visual neuroscience and their lifetime contributions continue to impact and guide research within the Dalhousie community.

T43: Freud’s Rat Man And The Meaning Of The Rat Torture**Authors:** John Barresi

Abstract: Freud’s 1909 case of obsessional neurosis of a patient called the Rat Man is unique in being the only case for which he saved his process notes. I provide a novel interpretation to the ‘rat torture’, which is the signature obsession in this case. This torture involves rats entering the anus of a prone victim from a pot. Freud provided an elaborate account of why the patient feared that this punishment would be given to his lady love and dead father. I propose, instead, that the patient’s father engaged in anal sexual abuse with his children and provide evidence from the process notes that supports this hypothesis. I point out how this explanation is more congruent with Freud’s original “seduction” theory than to his later “fantasy” theory of the basis for neuroses, and relate these two theories to recent debates over the possibility of recovered memories of childhood sexual abuse.

Posters

P1: Drinking to keep pace: Testing the moderating influences of sensation seeking and extroversion on alcohol consumption in drinking buddy dyads

Authors: Raquel Nogueira-Arjona, Trevor CJ Shannon, Simon B Sherry Ivy-Lee Kehayes, Sherry H Stewart

Abstract: Peterson et al. (2005) showed that modelling contributes to college students’ alcohol use, with highly extroverted males being most susceptible to matching strangers’ drinking levels in a lab-based context. We examined if: a) there was a significant relationship between target and buddy drinking levels in college students’ dyadic drinking situations; b) both target extraversion and sensation-seeking positively moderate this target-buddy drinking levels relationship. Data were collected from 149 targets (74% F) and their same-sex drinking buddy. The target completed personality measures, and both completed a past 30-day Timeline Follow Back examining drinking levels and context. Linear mixed-effects modeling confirmed the first social matching hypothesis, and revealed that extraversion, but not sensation-seeking, positively moderated the relationship between buddy and target daily dyadic drinking levels. Findings extend previous research to a real-world context, modeling of a buddy’s drinking levels, and to women. Results may guide modifications to personality-based interventions for college drinking.

P2: Specificity of Four Personality Correlates of Concurrent Substance Use Among Opiate Substitution Therapy Clients

Authors: Ioan Mahu, Sean Barrett, Patricia Conrod, Mike Lawrence, Annie Chinneck, Julie Bruneau, Sherry Stewart.

Abstract: The leading treatment for opioid addiction in Canada is methadone maintenance therapy (MMT). While efficacious for treating opiate addiction, MMT fails to address patients' concurrent polysubstance use. This study was designed to validate the Substance Use Risk Profile Scale (SURPS; Woicik et al., 2009) in a MMT sample, and to demonstrate theoretically-relevant relationships of personality to recent substance use. One-hundred-thirty-eight MMT patients from four clinics in Quebec and Nova Scotia participated. Bayesian confirmatory factor analysis established the structural validity of the four-factor personality risk model in this population. SURPS subscales were then used as predictors for specific forms of recent substance use. Hopelessness predicted past 30-day opiate and tranquilizer use, anxiety sensitivity predicted past 30-day tranquilizer use, and sensation seeking predicted past 30-day stimulant, alcohol, and marijuana use. Impulsivity predicted past 30-day injection drug use (particularly stimulant injection). Thus, personality influences substance use patterns, even at latter stages of addiction.

P3: Do I Want Babies Now? Reproductive Investment of *Gryllus texensis* under the Interaction of Ageing and Immune Challenge.

Authors: Marco Lee, Ann Johnsen Co-supervisors: Shelley Adamo, Atsushi Miyashita

Abstract: To maintain maximum fitness in a variable environment, individuals must adapt to environmental cues by altering investments into current reproduction, somatic maintenance and future reproduction. The terminal investment hypothesis predicts that as prospects of future reproduction decrease, investment into reproduction should increase. However, the threshold for terminal investment may shift with age and disease risk. Interactions between immune challenge (IC) and age were examined by monitoring lifetime reproductive investments of female crickets that were exposed to an IC at different ages. We predicted that younger females would over-invest in immunity relative to older females because younger females will lose proportionately more of their total reproductive output if they die of disease than will older females who have already laid many of their eggs. We also predicted that the increased investment in immunity would lead to younger females having lower current reproduction compared with older females. Regardless of treatment, older females laid more eggs over a 7-day period than did young females. Contrary to our prediction, phenoloxidase, an immune enzyme, showed greater activity in older females.

P4: An ERP study of reading ability and polyunsaturated fatty acid consumption in childhood

Authors: Clara J Lownie, Alena Galilee, Aaron J Newman

Abstract: Diet is one of many environmental factors influencing a child's ability to acquire proficient reading skills. Polyunsaturated fatty acids (PUFA) are a group of nutrients that are critical to the neurodevelopment of several cognitive abilities, such as reading. This study used electroencephalography to record electrical activity in the brain following visual word stimuli within a reading task. Specifically, this research focuses on one event-related potential component, the N170, associated with two reading related phenomena; print tuning and lexical tuning. The amplitude of the N170, behavioural measures of reading, and estimated consumption of PUFA in children grades 2 to 6 in the Halifax Regional Municipality were analyzed. While the findings of this exploratory study do not provide conclusive evidence to support the hypothesis that estimated PUFA consumption is correlated to print or lexical tuning, these results suggest that print and lexical tuning effects may actually be negatively correlated to PUFA consumption.

P5: Investigating the Path from Childhood Maltreatment to Alcohol Problems in a Sample Child Welfare-Involved Youth**Authors:** Tristan Park, Sherry H Stewart, Mohammed Al-Hamdani, Kara Thompson, Christine Wekerle**Abstract:** PTSD symptoms and drinking motives have both been identified as mediators in the relationship between trauma and alcohol misuse. We sought to extend recent research suggesting that both PTSD symptoms and, in turn, drinking motives mediate the relationship between childhood maltreatment and alcohol misuse. Five-hundred and sixty-four adolescents (54% female; mean [SD] age = 15.9 [1.1] years) in the care of child welfare services completed validated measures of childhood maltreatment, PTSD symptoms, drinking motives, and alcohol misuse as part of the Maltreatment Adolescent Pathways longitudinal study. Results supported a chained mediation model where childhood maltreatment was associated with elevated PTSD symptoms, which in turn predicted higher coping motives, which was associated with higher alcohol misuse. Single mediator models for PTSD symptoms and coping motives were not significant. The results suggest that both PTSD symptoms and coping motives are necessary to explain the relationship between childhood maltreatment and alcohol misuse.**P6:** Self-compassion as a resilience factor for alcohol-involved sexual assault and negative emotional outcomes.**Author:** Noelle Strickland**Abstract:** Experiencing alcohol-involved sexual assault (AISA) may result in self-blame, while self-compassion (SC; extending kindness towards oneself) may promote resilience from the negative emotional consequences of AISA by attenuating and/or counteracting the association between AISA and negative emotional outcomes. Undergraduates ($N = 785$) completed measures tapping past-term AISA (Kehayes et al., in press), SC (i.e., Self-Compassion Scale; SCS; Neff, 2003), and anxiety and depression (K10; Kessler et al., 2002). The SCS was scored as two higher-order domains (SC, self-criticism) and six lower-order facets (self-kindness, mindfulness, common humanity, over-identification, self-judgement, isolation). Rather than attenuation, the higher-order SCS domains showed main effects: high SC and low self-criticism counteracted the adverse effects of AISA on anxiety and depression. Similarly, the lower-order SCS facets of high self-kindness and low over-identification counteracted the adverse effects of AISA on anxiety/depression – with the additional SCS facets of low self-judgement and isolation counteracting the effect of AISA on depression.**P7:** What Influences Children's Sensitivity to Violations of Authority as a Moral Foundation?**Authors:** Emily Puddicombe, Robin Curtis, Chris Moore**Abstract:** According to Moral Foundations Theory (Haidt, 2012), human beings are innately sensitive to five core moral domains, which can be influenced by upbringing and culture. One of these foundations is authority. However, limited research has investigated children's sensitivity to authority as a moral foundation. Studies have found that children's age, potential punishments, and authoritative figure's status influence children's response towards authority. The current study used the Moral Foundation Questionnaire for Kids to measure 31 children's sensitivity to authority when presented with six vignettes depicting violations of authority towards teachers and parents. Badness ratings and punishment choices were taken for each vignette. A mixed two by two ANOVA revealed no main effect of age on children's badness ratings and no main effect of authoritative figure type on children's overall badness ratings. However, an independent samples t-test found older children were significantly more punishing than younger children. Limitations and implications are discussed.

P8: Changes in Anxious Mood, Stress, and Cannabis Use Across the Menstrual Cycle: A Daily Diary Study**Authors:** N Hanzal, KM Joyce, SH Stewart

Abstract: Addictions research has predominantly focused on substance abuse in male-dominated research samples, indicating that findings may not generalize to women. Menstrual cycle (MC) phase appears to influence changes in addictive/risky behaviour. The current study examined phase-specific changes in anxious mood, stress, and cannabis use quantity across the MC. Consistent with the self-medication hypothesis (SMH), it was predicted that the quantity of cannabis used would increase during the (pre-)menstrual phases relative to other MC phases. It was also predicted that this increase in quantity of cannabis use would be associated with an increase in stress levels and anxious affect. For the current study, data were obtained from 14 female cannabis users using a 32-day daily diary method, with questions pertaining to anxious mood, stress, and quantity of cannabis used. Findings suggest that the quantity of cannabis used is phase-specific, while stress nor anxious affect are phase-specific variables.

P9: Alertness and Human Performance**Authors:** Lindsey Puddicombe, Bronwyn O'Connor, Samantha Howard, Piper Sawchyn
(Supervisors: Dr. Klein, Colin McCormick & Austin Hurst)

Abstract: Alerting is an attentional mechanism that increases arousal and helps us prepare to make responses. Human performance can be measured using reaction time and accuracy. A prior study by Posner et. al(1973) looked at the relationship between reaction time and accuracy when alerting at different time intervals. They found when reaction time was fastest; error rate was highest, which is a speed- accuracy trade-off (SAT). Our follow-up experiment required participants to indicate whether targets appeared on the left or right side, using either compatible or incompatible responses. Our manipulation was our method for alerting: participants would hear either an exogenous or endogenous warning signal. The goal of our study was to see if these different signal types had differing effects on human performance. The interval duration and the compatibility cues had a significant effect on both reaction time and accuracy. The differences between endogenous and exogenous signals are further considered.

P10: Investigating the Effect of Visual Noise on Cognitive Load and Memory using Item-Method Directed Forgetting**Authors:** Jessie Pappin, Tracy Taylor (supervisor)

Abstract: The present study investigated the effect of visual noise induced cognitive load on item-method directed forgetting. Study words were presented to participants superimposed on one of three levels of visual noise. Study words were followed by a memory instruction to remember or forget the previously seen word. Participants' completed a recognition test for both remember and forget words. The directed forgetting effect, defined as better recognition of remember words than of forget words, was seen in all levels of visual noise. Recognition accuracy was significantly lower in the high noise condition than in both the medium and low condition. These findings suggest that high levels of visual noise can facilitate item-method directed forgetting.

P11: Prism Adaptation in Older Adults: an ERP Analysis**Authors:** Zoë Lazar-Kurz, Gail Eskes, Aaron Newman, Stephan MacLean, Katie MacKearney, Jasmine Aziz, Cindy Hamon-Hill

Abstract: In this study, we measured prism adaptation (PA) in older adults as a marker for neuroplasticity. Currently, little is known about the neural correlates of PA, and even less in older adults. Event related potentials (ERPs) were recorded in response to a target reaching task while wearing prism or clear lenses. Past research in young adults suggested two ERP components facilitate PA: the P300 and the feedback error related negativity (fERN; MacLean et al., 2015). In our study of

older adults, while behavioural tests converged with young adults, only the fERN was elicited, not the P300. As seen in young adults, the fERN amplitude was related to spatial accuracy of the pointing in relation to the location of the target. These preliminary results may suggest the neural mechanisms associated with PA differ in older and younger adults. However, these results need to be verified by directly comparing younger participants with the same procedure.

P12: *In vivo* application of the photoswitch BENAQ in a rat model of acquired retinal degeneration

Autho: Heather Gerrie

Abstract: Millions of people worldwide suffer from degenerative retinal diseases, such as retinitis pigmentosa (RP) and age-related macular degeneration (AMD). There are currently no therapies available to reverse vision loss, leaving individuals with the prospect of irreversible blindness. AMD and RP specifically target the photoreceptor cells of the retina and induce progressive deterioration. As the primary light sensitive cells of the retina, photoreceptor loss results in decreased visual acuity and severe visual impairment. To reduce the consequences of retinal degeneration, a series of azobenzene photoswitch drugs have been developed as a potential therapeutic strategy. Photoswitches are able to confer light sensitivity onto non-light responsive cells, such as retinal ganglion cells, causing them to fire in response to light stimulus. However, the effects of photoswitches have only been demonstrated in congenital models of photoreceptor degeneration, despite the acquired nature of some retinal diseases, including AMD. Additionally, it is not well understood how the visual cortex responds when a photoswitch is applied to the retina. Therefore, this project sought to test cortical response to photoswitch application in a model of acquired retinal degeneration.

P13: Investigating the ability of superoxide to pass dendritic spine plasma membranes in hippocampal neurons via chloride channels using fluorescence imaging.

Authors: Hannah MacNeil, Alan Fine

Abstract: Reactive oxygen species such as superoxide have been shown to carry out individual processes involved in long-term potentiation (LTP). Superoxide in neurons is produced extracellularly and until recently, it was assumed that it could not pass through the membrane. Recent findings suggest that superoxide can pass membranes in epithelial cells via chloride channels. I hypothesized that a similar mechanism occurs in hippocampal neurons, allowing superoxide to perform distinct functions in intracellular signalling involved in LTP. The current study tested the hypothesis that superoxide can pass through the plasma membrane in spines of rat hippocampal neurons by detecting superoxide with fluorescent dye and imaging the cells before and after superoxide production. The trend of the results supports the original hypothesis, as a significant increase in fluorescence was observed in some putative spines 10 seconds after superoxide production, and this increase was eliminated when chloride channels were blocked.

P14: The unresolved role of the cell adhesion molecule family, neuroligins, and activity on the stability of synapses between hippocampal neurons

Authors: Sydney Harris, Annette Kolar, Dylan Quinn, Stefan Krueger

Abstract: Synapse elimination and formation is a necessary part of post-natal development. It optimizes the wiring of neuronal circuits, ensuring proper processing of sensory input and behavioural output. The process of synaptic refinement is regulated by neural activity, which affects the structural and functional integrity of synapses through possibly global homeostatic mechanisms. Besides activity, cell adhesion molecules critically influence synapse refinement. Neuroligins, a family of cell adhesion molecules, previously implicated in modulation of synaptic transmission, have been shown to stabilize synapses. It is presently unclear whether neuroligins affect synapse refinement structurally or by modulating synaptic transmission. To address this, we inhibited AMPA-type and NMDA-type glutamate receptors in hippocampal neurons in the presence and absence of neuroligin

knockdowns. Although chronic blockade of glutamate receptors led to a decrease in eliminated synapses, the density was unaltered. Moreover, only one of two neurexin knockdowns caused significant decrease in synapse density and significant increase in synapse elimination in a glutamate receptor blockade, preventing us from providing a conclusive result.

P15: Behavioural phenotypes of the MDGA2+/- mouse model of autism spectrum disorder (ASD)

Authors: Michaela K. Purdon, Emre Fertan, Richard E. Brown

Abstract: A MAM domain-containing glycosylphosphatidylinositol anchor 2 (*MDGA2*) gene has been linked to ASD in humans [Bucan M. et al., 2009, *PLOS*. 5:6]. Connor S. A. et al. [2016, *Neuron* 91:5, 1052-1068] showed that haploinsufficiency in the MDGA2+/- mouse model produced behavioural phenotypes paralleling those seen in humans with ASD. The MDGA2+/- mice used by Connor et al. (2016) were bred using a C57BL/6N background which has an rd8 mutation that causes retinal spotting. We used 2-month old transgenic (MDGA2+/-) mice and wildtype (MDGA2+/+) littermate controls on a C57BL/6J background to determine whether visual impairment was a confounding variable in the Connor et al. (2016) study. Results show differences on tests of sociability, motor ability, and cognition, indicating visual impairments and/or background strain may influence the behavioural phenotype of this model.

P16: Effects of Atrx haploinsufficiency and mother-pup interactions on cognitive, social and emotional behaviours in adulthood

Authors: Kristen Lee, Donna Goguen and Ian Weaver

Abstract: A child's early-life conditions both before birth and in infancy has profound effects on long-term physical and mental health outcomes of an individual. The relationship between the quality of early environment and long-term developmental programming appears to be mediated, in part, by the degree of positive attachment in parent-infant bonding and parental investment during early life. We have previously shown that offspring of low maternal care mothers have reduced forebrain Atrx levels. Children born with mutations in the Atrx gene have intellectual disability and autistic-like behavior. Using wildtype and Atrx heterozygous C57/B6 mice, we examined the role of ATRX and prenatal stress in mitigating the effects of variations in maternal behavior during early brain development and social emotional and cognitive behaviors into adulthood. Our results suggest that prenatally stressed offspring have social and behavioral deficits associated with reduced Atrx levels.

P17: Using the Hebb Legacy to Integrate Psychological Science

Authors: Raymond M. Klein, Richard E. Brown

Abstract: The birth of our relatively young science has been described (Boring, 1950) as the product of a marriage between physiology and philosophy. Whereas we might be described as an incredibly successful offspring we are presently experiencing something of a disintegrating identity crisis. As proposed by Posner & Rothbart (2004, 2007) the framework advocated by D. O. Hebb (1949), in the "Organization of Behavior: A Neuropsychological Theory", and other works, can provide the foundation for the integration of our discipline and the framework for averting this crisis. Hebb (1949) proposed that synaptic change is the basis of neural activity (which we now call the "Hebb synapse") and developed the idea of a "cell assembly" or neural network to account for the ability of the brain to retain information in the absence of sensory stimulation. Hebb sought to explain psychological concepts, from learning and memory to brain development, motivation, emotion and brain disorders in neuroscientific terms. As noted by Posner & Rothbart, technological advances in neuroimaging, genetics and computational modeling, offer a firm and realistic foundation for research and training aimed at understanding how environment and heredity interact to shape the neural networks that underlie the organization and disorganization of behaviour, cognition and affect.